CS5730 Operating Systems Midterm (2021/11/30)

Name:_____
ID#:_____

Essay Questions: Your answers should be as concise and clear as possible

1 (5 pt) Ordinarily the `exec()` system call follows the `fork()`. Explain what would happen if a programmer were to inadvertently place the call to `exec()` before the call to `fork()`.

2 (5 pt) Why does the socket communication use the port number instead of process id number to identify server or client process?

3 (8 pt) What is processor affinity? Explain why it needed to be considered during process scheduling on multiple CPU systems.

4 (7 pt) Explain why implementing synchronization primitives by disabling interrupts is not appropriate in a single-processor system if the synchronization primitives are to be used in user-level programs.

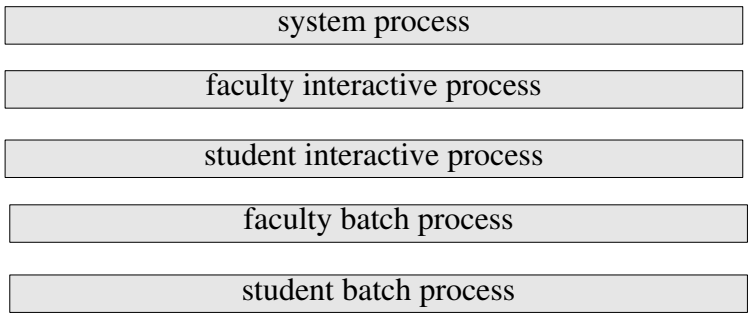
5 (15 pt) Discuss how the following pairs of scheduling criteria conflict in certain settings.

5.1 CPU utilization and response time
5.2 Average turnaround time and maximum waiting time
5.3 I/O device utilization and CPU utilization
5.4 Deadline and fairness
5.5 Throughput and context switch overhead

6 Multilevel queue scheduling

6.1 (5 pt) List two possible scheduling among queues, which we discussed in class.

6.2 (10 pt) Modify the following multilevel queue scheduling system such that a new process will automatically enter the appropriate queue after a few runs.



7 Multilevel queue scheduling

7.1 (5 pt) What is **busy-waiting** in the discussion of critical section problem? Explain how **busy-waiting** may degrade the system performance.

7.2 (10 pt) How to implement semaphore without **busy-waiting** using `block` and `wakeup`? You need to write down the definition of data structure `semaphore` and `wait` and `signal` operations.

8 (10 pt) We have discussed the correct solution to the consumer-producer problem, which does not use any synchronization mechanism, as shown below. Please explain why it can only use `BUFFER_SIZE-1` elements of buffer.

Producer:

```
while (true) {
      /* produce an item in next produced */
      while (((in + 1) % BUFFER_SIZE) == out)
            ; /* do nothing */
      buffer[in] = next_produced;
      in = (in + 1) % BUFFER_SIZE;
}
```

Consumer:

```
while (true) {
      while (in == out)
            ; /* do nothing */
      next_consumed = buffer[out];
      out = (out + 1) % BUFFER_SIZE;

      /* consume the item in next consumed */
}
```

9 Consider two processes, $\mathbf{P_1}$ and $\mathbf{P_2}$, where $p_1 = d_1 = 50$, $t_1 = 25$, $p_2 = d_2 = 75$, and $t_2 = 30$.

9.1 (10 pt) Can these two processes be scheduled using rate-monotonic scheduling (RMS)? Illustrate your answer using a Gantt chart such as the ones we did in class.

9.2 (10 pt) Illustrate the scheduling of these two processes using earliest-deadline-first (EDF) scheduling.